

CodeBox Coding Platform

Final Development Flow (MERN + PostgreSQL + Judge Microservice)

System Architecture

Frontend (React) → CodeBox Backend (Node + Express) → Redis Queue → Judge Service (Docker Worker) →
Shared Testcase Storage → PostgreSQL Database

PHASE 0 — System Setup

Install: - Docker - PostgreSQL - Redis - Node.js Create project structure: CodeBox/ frontend/ backend/ judge/ testcases/ docker-compose.yml

PHASE 1 — Judge Service

Goal: Execute code inside Docker safely Step 1: Build CLI judge prototype that: - takes code - takes language - takes input file - runs inside docker - returns output Step 2: Support multiple testcases Judge loads testcases from: /testcases/{problemId}/input1.txt Step 3: Add Express API: POST /judge submissionId problemId code language

PHASE 2 — Shared Testcase Storage

Create structure: testcases/ 1/ input1.txt output1.txt input2.txt output2.txt Mount this folder inside: - backend container - judge container Judge loads locally. Backend never sends testcase files.

PHASE 3 — Database (PostgreSQL)

Tables: Users Problems Testcases (metadata only) Submissions Submission Results

PHASE 4 — Backend (CodeBox API)

Build APIs: Auth: POST /signup POST /login Problems: GET /problems GET /problems/:id Submission: POST /submit Flow: Save submission(status=pending) Push job to Redis queue

PHASE 5 — Redis Queue

Install BullMQ Submission flow: User submits → Backend saves submission → Adds job to Redis → Judge worker picks job → Executes code → Updates DB

PHASE 6 — Judge Worker Mode

Judge: - listens to Redis - picks submission jobs - connects to DB - executes code - updates DB

PHASE 7 — Frontend

Build UI: - Login/signup - Problem list - Problem description - Monaco editor - Submit button - Submission history Submission should: Show 'Pending' Poll backend Update result

PHASE 8 — Optional Enhancements

- WebSockets for live verdict - Leaderboard - Contest system - Custom input run feature - Admin problem upload panel

Submission Lifecycle

User clicks Submit Frontend sends code Backend stores submission Backend pushes Redis job Judge worker picks job Judge fetches submission from DB Judge loads local testcases Runs in Docker sandbox Compares output Updates PostgreSQL Frontend fetches result

MVP Completion Checklist

Feature	Status
Docker execution	
Multiple testcases	
Judge API	
PostgreSQL schema	
Backend submission	
Redis queue	
Async judging	
Frontend editor	
Submission history	